

SUMMARY

Instructional Designer and STEM educator with over 5 years of experience designing curriculum, developing digital learning materials, and integrating technology in education. Skilled in applying instructional design models to create learner-centered experiences. Proficient in multiple programming languages and design tools. Passionate about expanding access to quality education through innovative digital pedagogy.

Experience

Fundi Robotics Limited

Technical Associate - (Fundi@Work)

Jan 2025 – Present

Kampala, Uganda

- Implemented a specialized robotics curriculum tailored to bridge the skill gap for 20 university interns and 22 out-of-school improving job readiness for participants through hands-on technical instruction.
- Conducted Training of primary and secondary Teachers aimed towards tool integration in the classroom.
- Designed the Logic Gates teaching aid to simplify learning in Senior 4 for the Enhanced Science Curriculum.
- Designed and wrote learning modules for the Robotics and Coding Curricula.
- Participated in a 1 week Design Thinking Course facilitated by Design Without Borders
- Maintained, troubleshoot and repaired robotics equipment, 3D printers, sensors and electronic tools to ensure uninterrupted, safe and effective learning environments.

Fundi Robotics Limited

Teaching Associate - Robotics (Fundi@School)

Jan 2022 – Dec 2024

Kampala, Uganda

- Developed hands-on learning aids, such as models of the digestive system, reproductive organs, electric bell to enhance science teaching at primary schools.
- Conducted Training of primary and secondary Teachers aimed towards tool integration in the classroom.
- Conducted interactive Robotics training and Computer programming sessions at schools.
- Designed and wrote learning modules for the Robotics, Coding and Enhanced Science Curricula.
- Integrated digital tools (Scratch, PictoBlox, Arduino, Tinkercad, Google Meet, Google classroom) to enhance computational thinking and problem-solving skills of students.
- Provided one-on-one support to students challenged by advanced science concepts.
- Participated in the monitoring and evaluation of the progress of children in learning their centers.

Fundi Robotics Limited

Rollout Support Staff (Fundi@School)

Jan 2020 – Dec 2021

Kampala, Uganda

- Provided in-class support to teachers, assisting with lesson delivery, and enhancing the quality of learning for children
- Conducted evaluations with students and teachers to assess learning progress and the effectiveness of teaching methods.
- Assembled and distributed learning materials to schools to enhance science learning
- Edited Teachers and Learners Guide for the Enhanced Science Curriculum
- Maintained an accurate inventory of learning materials, ensuring resources were available and appropriately utilized for the intended purposes

Martin Luther King College

Teacher of Physics and Mathematics

Sept. 2019 – Dec 2021

Kireka, Uganda

- › Served as the Class Teacher of Senior Six - Sciences
- › Designed Schemes of Work out of the National Curriculum and innovative lesson plans to deliver Mathematics and Physics lessons
- › Assessed and kept records of student progress through tests, assignments and practical exercises.
- › Participated in professional development to stay up to date with advances in the education of Physics, Mathematics and STEM.

St Henry's College - Namugongo

Teacher of Physics and Mathematics - School Practice

June 2019 - Aug 2019

Namugongo - Uganda

- › Gained hands-on experience in lesson planning, delivery, and assessment.
- › Worked closely with mentor teachers to refine instructional strategies and student engagement techniques.
- › Created detailed lesson plans aligned with the national curriculum for Physics and Mathematics.
- › Learned effective teaching strategies by observing experienced Physics and Math teachers.
- › Collaborated with other subject teachers to improve learning experience for students.

St Gerald's College - Nyakibale

Teacher of Physics and Mathematics - School Practice

June 2018 - Aug 2018

Rukungiri - Uganda

- › Gained hands-on experience in lesson planning, delivery, and assessment.
- › Worked closely with mentor teachers to refine instructional strategies and student engagement techniques.
- › Created detailed lesson plans aligned with the national curriculum for Physics and Mathematics.
- › Learned effective teaching strategies by observing experienced Physics and Math teachers.
- › Collaborated with other subject teachers to improve learning experience for students.

Education

Mbarara University of Science and Technology

Bachelor of Science with Education - Physical

2016 – 2019

Mbarara , Uganda

Luzira C.O.U Senior Secondary School

Uganda Advanced Certificate of Education

2014 – 2015

Kampala , Uganda

Luzira C.O.U Senior Secondary School

Uganda Certificate of Education

2010 – 2013

Kampala , Uganda

Luzira C.O.U Primary School

Primary Leaving Examination

2009

Kampala , Uganda

Technical Skills

LMS Google Classroom, Moodle (self-taught), Zoom, Google Meet, Microsoft Teams, Asana, Edx

Computer Programming Python, Django, JavaScript, PictoBlox, Arduino, HTML, CSS

Computer Aided Design SolidWorks, FreeCAD, Blender, TinkerCAD, Canva, $\text{\LaTeX}$

Microsoft Office Suite MS Word, Excel, PowerPoint

Multimedia and Visual Design DaVinci Resolve, VSDC, Adobe Photoshop

Soft Skills

Traits Hardworking, Time management, Collaboration, Kaizen, Adaptability, Curiosity

Hobbies Soccer, Biking, Content Creation, Music, Movies, Traveling

Extra-Work Activities

Capacity Building and Professional Development

2024 - Present

- Learning Artificial Intelligence and Machine learning (Self-taught through various MOOCs.
- Exploring Unity, Blender and Virtual Reality Design to develop learning models that allow learners to conceptualize advanced science concepts. I designed a Digestive System Model as first step toward VR-based science learning, the goal is to enable learners explore the process of digestion from Mouth to Anus using VR glasses.

Amigos Football Club (Kampala - Uganda)

2024 - Present

- As the captain of the club, I Organize and lead weekly training sessions and matches for a group of friends focused on physical fitness and team cohesion. Promote a healthy, active lifestyle while fostering teamwork, discipline, and sportsmanship.

STEM Club (Python and Robotics Training) (Kireka - Uganda)

2023 - Present

- I design and facilitate 3-week intensive training programs in Python Programming and Robotics for secondary students during school holidays that take absolute beginners to building functional projects (GUI applications, robotic systems) in 15 days

STEM Voices Uganda (YouTube channel)

2025-Present

- STEM Voices Uganda is a consortium of dedicated educators who believe in the power of science as a tool for sustainable development. This podcast highlights how science education drives innovation, fosters critical thinking, and contributes to the growth of Uganda and beyond. I edit the videos and participate in the content creation of the topics to be discussed.

Certificates

Strategies for Personal Growth and Development | Young African Leaders Initiative | May 2025

Asana Workflow Specialist | Asana | May 2025

Finalist | International Youth Mathematics Competition | November 2019

135 International course on Astronomy NASE-IAU | Network for Astronomy School Education | April 2018

Class President | Mbarara University B.Sc(Ed) -Physics Majors | June 2018

Sports Participant | Mbarara University of Science | June 2018

References

1. **Obote Geoffrey Tale**

Senior Associate, Fundi Robotics Limited , Kampala, Uganda

Phone : +256-777-808127

Email: geoffreytale@gmail.com

2. **Dr Ndugu Nelson**

Lecturer, Makerere University, Kampala, Uganda

Phone : +256-775-177759

Email: ndugu.nelson89@gmail.com

3. **Philemon Mindra**

Hardware and Storage Engineer, CNOOC Limited, Kampala, Uganda

Phone : +256-776-798363

Email: philemon.mindra@outlook.com